Worksheet 31 · Circle Theorems IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Recall and apply the circle theorems • Find missing angles in circle diagrams • Give reasons using correct circle theorem language		
Grade: / 57	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Circle Theorems*

Question 1 (3 marks)

In the diagram below, P and Q are points on a circle with centre O.

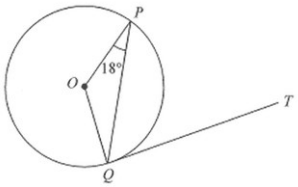


Diagram **NOT** accurately drawn

QT is a tangent to the circle.
Angle $OPQ = 18°$

Work out the size of angle PQT.
Give a reason for each stage of your working.

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Question 2 (4 marks)

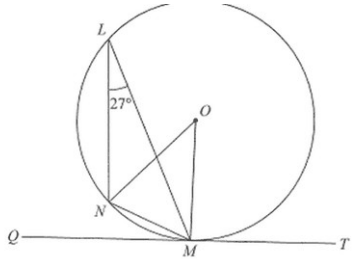


Diagram **NOT** accurately drawn

, M and N are points on a circle, centre O.
MT is the tangent to the circle at M.

) (i) Find the size of angle NOM.

°

(ii) Give a reason for your answer.

(2)

) (i) Find the size of angle NMQ.

°

(ii) Give a reason for your answer.

(2)

Question 3 (4 marks)

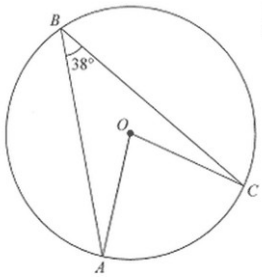


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle $ABC = 38°$

Work out the size of angle OAC.
Give a reason for each stage of your working.

°

Question 4 (5 marks)

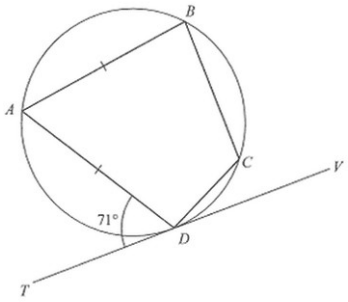


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
TDV is the tangent to the circle at D.

$AB = AD$
Angle $ADT = 71°$

Work out the size of angle BCD.
Give a reason for each stage of your working.

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Question 5 (2 marks)

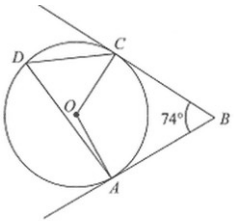


Diagram **NOT** accurately drawn

A, *C* and *D* are points on a circle, centre *O*.
AB and *CB* are tangents to the circle.

Angle *ABC* = 74°

Work out the size of angle *ADC*.
Show your working clearly.

.....°

Question 6 (4 marks)

A, *B*, *C* and *D* are points on a circle, centre *O*.

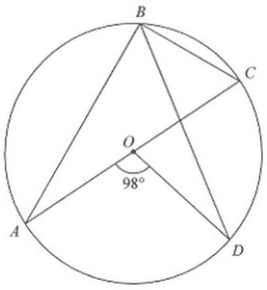


Diagram **NOT** accurately drawn

AOC is a diameter of the circle.

Angle *AOD* = 98°

Work out the size of angle *DBC*.
Give a reason for each stage in your working.

.....°

Question 7 (4 marks)

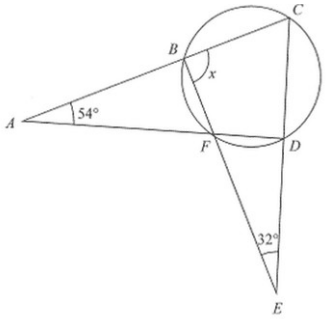


Diagram **NOT** accurately drawn

B, *C*, *D* and *F* are points on a circle.
ABC, *AFD*, *BFE* and *CDE* are straight lines.

Work out the size of angle *x*.
Show your working clearly.

x = ...°

Question 8 (5 marks)

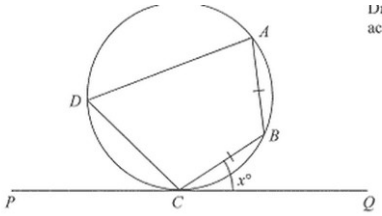


Diagram **NOT** accurately drawn

A, *B*, *C* and *D* are points on a circle.
PCQ is a tangent to the circle.
AB = *CB*.

Angle *BCQ* = *x*°

Prove that angle *CDA* = 2*x*°
Give reasons for each stage in your working.

Question 9 (3 marks)

P, *Q*, *R*, *S* and *T* are points on a circle with centre *O*.

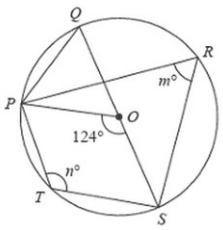


Diagram **NOT** accurately drawn

QOS is a diameter of the circle.

angle *POS* = 124° angle *PRS* = *m*° angle *PTS* = *n*°

(a) Find the value of

(i) *m*

.....

(ii) *n*

.....

(2)

(b) Find the size of angle *QPO*.

.....°

(1)

Question 10 (5 marks)

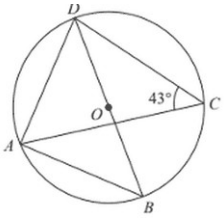


Diagram **NOT** accurately drawn

A, *B*, *C* and *D* are points on a circle, centre *O*.
DOB is a diameter of the circle.
Angle *ACD* = 43°

Work out the size of angle *ADB*.
Give a reason for each stage in your working.

.....°

Question 11 (3 marks)

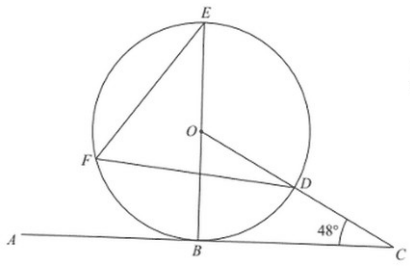


Diagram **NOT** accurately drawn

B, *D*, *E* and *F* are points on a circle, centre *O*.
ABC is a tangent to the circle.
ODC is a straight line.

BOE is a diameter of the circle.

Angle *BCD* = 48°

Find the size of angle *DFE*.

.....°

Question 12 (4 marks)

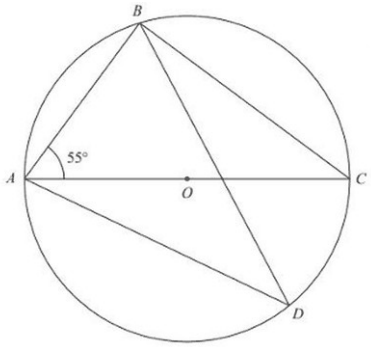


Diagram **NOT** accurately drawn

A, *B*, *C* and *D* are points on a circle, centre *O*
AOC is a diameter of the circle.

Angle *BAC* = 55°

Work out the size of angle *ADB*
Give a reason for each stage of your working.

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Question 13 (3 marks)

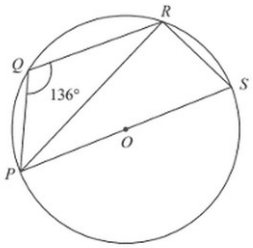


Diagram **NOT** accurately drawn

P, *Q*, *R* and *S* are points on a circle with centre *O*

PS is a diameter of the circle.

Angle *PQR* = 136°

Work out the size of angle *RPS*

.....°

Question 14 (4 marks)

P, *Q* and *R* are points on a circle, centre *O*.
TRV is the tangent to the circle at *R*.

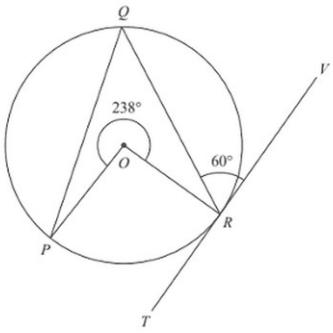


Diagram **NOT** accurately drawn

Reflex angle *POR* = 238°
Angle *QRV* = 60°

Calculate the size of angle *OPQ*.
Give a reason for each stage of your working.

.....°

Question 15 (4 marks)

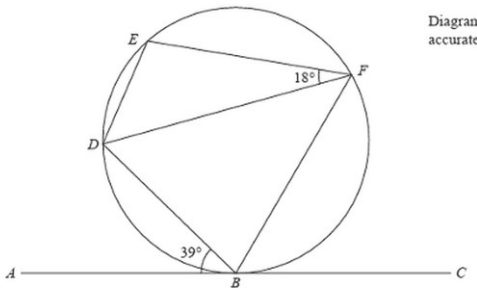


Diagram **NOT** accurately drawn

B, *D*, *E* and *F* are points on a circle.

ABC is the tangent at *B* to the circle.
Angle *ABD* = 39°
Angle *EFD* = 18°

Work out the size of angle *BDE*.
Give reasons for your working.

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